

Spectral Analysis

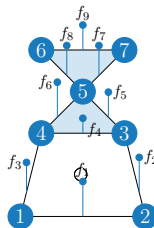
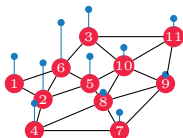
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Intelligent Systems Department

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Premises

- We processed topological data on the **vertex/simplex/cell** domain
 - ⇒ Given a **structure** and some **signal features**
 - ⇒ **Build** local models to process these data
 - ⇒ **Now** get insights from a spectral perspective
 - ⇒ **Ex.** Graph spectral theory is key to understanding networks or dynamics on them

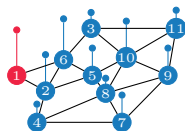


Graph Convolutional Filter

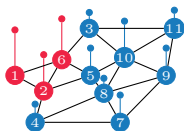
- Graph convolution $\Rightarrow$ Linear combination of shifted versions of the signal

$$\mathbf{z} = \sum_{k=0}^K h_k \mathbf{S}^k \mathbf{x} = \mathbf{H}(\mathbf{S}) \mathbf{x}$$

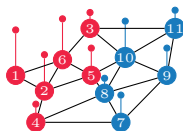
$\Rightarrow$ Linear combination of neighboring signal $\Rightarrow$ Local operation



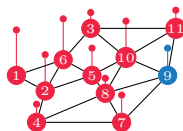
$\mathbf{Sx}$



$\mathbf{S}^2 \mathbf{x}$



$\mathbf{S}^3 \mathbf{x}$



$\mathbf{S}^4 \mathbf{x}$

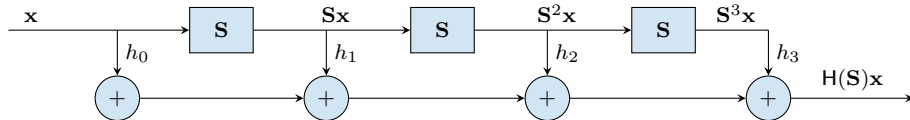


Image credit: F. Gama

Sum of Signals

► Superposition

- ⇒ Suppose we have a signal that is a sum of multiple frequencies.
- ⇒ How can we recover each frequency?

Sum of Signals

► Superposition

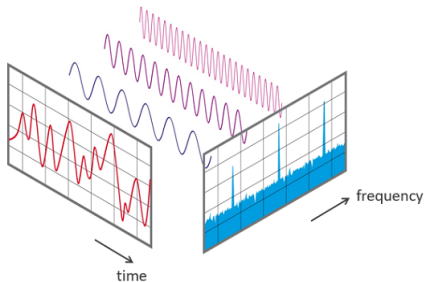
- ⇒ Suppose we have a signal that is a sum of multiple frequencies.
- ⇒ How can we recover each frequency?
- ⇒ **Answer:** the Fourier Transform



Fourier Transform

- **Decomposes** a signal into its constituent frequency components

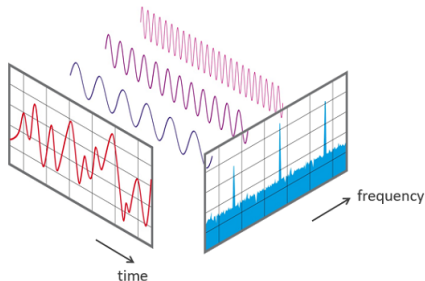
$$F(\omega) = \int_{-\infty}^{\infty} f(t) \cdot e^{-i\omega t} dt$$



Fourier Transform

- **Decomposes** a signal into its constituent frequency components

$$F(\omega) = \int_{-\infty}^{\infty} f(t) \cdot e^{-i\omega t} dt$$



- It reveals the **role** of each frequency component present in the signal

Outline: What Will You Learn Today?

- ▶ How to characterize topological models **and** data from a spectral perspective?

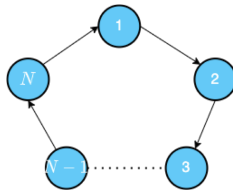
⇒ Spectral analysis of graph data

⇒ Spectral analysis beyond graphs

Spectral Analysis of Graph Data

Time as a Graph

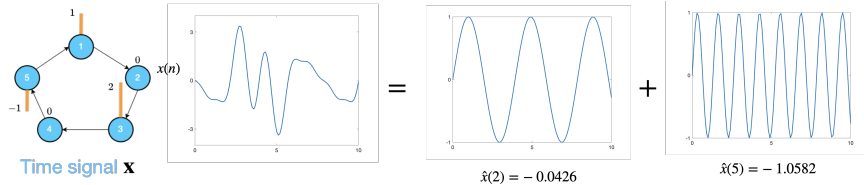
- Time as a graph of N nodes
⇒ Each node one time instant



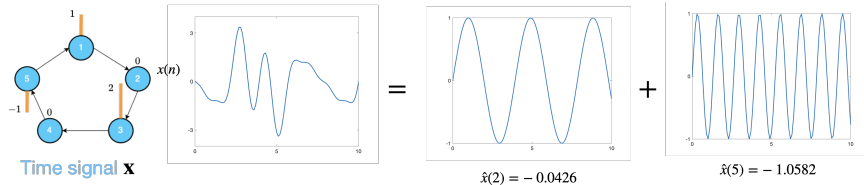
- **Shift operator:** Cyclic matrix

$$\mathbf{S} = \begin{pmatrix} 0 & 0 & 0 & \cdots & \cdots & 1 \\ 1 & 0 & 0 & \ddots & & 0 \\ 0 & 1 & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 & 0 \\ \vdots & & \ddots & 1 & 0 & 0 \\ 0 & \cdots & \cdots & 0 & 1 & 0 \end{pmatrix}.$$

Discrete Fourier Transform



Discrete Fourier Transform



- **Goal:** Do the same analysis for arbitrary graphs!

Eigendecomposition of Cyclic Matrices

► Eigen-decomposition $\mathbf{S} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}$

$$\boxed{\mathbf{S}} = \left(\begin{array}{c|c|c} | & & | \\ \mathbf{v}_1 & \dots & \mathbf{v}_N \\ | & & | \end{array} \right) \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_N \end{pmatrix} \left(\begin{array}{c|c|c} | & & | \\ \mathbf{v}_1 & \dots & \mathbf{v}_N \\ | & & | \end{array} \right)^{-1}$$

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- Eigenvectors:

$$\mathbf{V} = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & e^{\frac{j2\pi 1}{5}} & e^{\frac{j2\pi 2 \cdot 1}{5}} & e^{\frac{j2\pi 3 \cdot 1}{5}} & e^{\frac{j2\pi 4 \cdot 1}{5}} \\ 1 & e^{\frac{j2\pi 2}{5}} & e^{\frac{j2\pi 2 \cdot 2}{5}} & e^{\frac{j2\pi 3 \cdot 2}{5}} & e^{\frac{j2\pi 4 \cdot 2}{5}} \\ 1 & e^{\frac{j2\pi 3}{5}} & e^{\frac{j2\pi 2 \cdot 3}{5}} & e^{\frac{j2\pi 3 \cdot 3}{5}} & e^{\frac{j2\pi 4 \cdot 3}{5}} \\ 1 & e^{\frac{j2\pi 4}{5}} & e^{\frac{j2\pi 2 \cdot 4}{5}} & e^{\frac{j2\pi 3 \cdot 4}{5}} & e^{\frac{j2\pi 4 \cdot 4}{5}} \end{pmatrix}.$$

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- Eigenvectors:

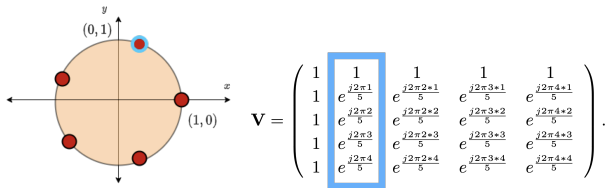
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- Eigenvalues:

$$\mathbf{\Lambda} = \text{diag}\left(1, e^{\frac{j2\pi}{5}}, e^{\frac{j2\pi 2}{5}}, e^{\frac{j2\pi 3}{5}}, e^{\frac{j2\pi 4}{5}}\right)$$

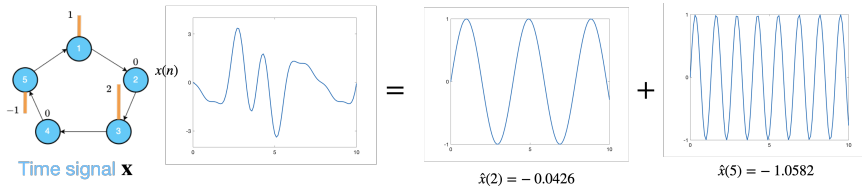
Eigendecomposition of Cyclic Matrices

- **Eigenvalue** λ_n : n th root of unity
 - ⇒ Each associated with an oscillation
 - ⇒ Temporal frequencies
- **Eigenvectors**: oscillation modes
 - ⇒ Complex exponentials: sines and cosines



Discrete Fourier Transform

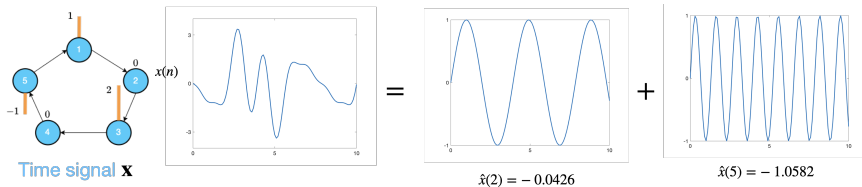
- **DFT**: express a temporal signal as a linear combination of complex exponentials (sine and cosine)



- **DFT** is a projection operation $\hat{\mathbf{x}} = \mathbf{V}^{-1}\mathbf{x}$ with $\hat{x}(k) = \sum_{n=0}^{N-1} x(n)e^{\frac{-j2\pi kn}{N}}$

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- **Inverse DFT**: $\mathbf{x} = \mathbf{V}\hat{\mathbf{x}}$
 $\Rightarrow$ Temporal signal = linear combination of spectral modes (complex exponentials)

Frequencies and Eigendecomposition

- **Frequencies** are located at the eigenvalues
 - ⇒ Frequencies are in an **increasing** order

$$\lambda_1 < \lambda_2 < \dots < \lambda_N$$

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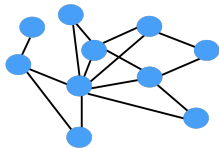
- **Eigenvectors** are associated to **increasing** frequencies
⇒ Indicate how much low-high varying components contribute to expressing the signal

$$\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3 \quad \mathbf{v}_4 \quad \mathbf{v}_5$$

$$\mathbf{V} = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & e^{\frac{j2\pi 1}{5}} & e^{\frac{j2\pi 2 \cdot 1}{5}} & e^{\frac{j2\pi 3 \cdot 1}{5}} & e^{\frac{j2\pi 4 \cdot 1}{5}} \\ 1 & e^{\frac{j2\pi 2}{5}} & e^{\frac{j2\pi 2 \cdot 2}{5}} & e^{\frac{j2\pi 3 \cdot 2}{5}} & e^{\frac{j2\pi 4 \cdot 2}{5}} \\ 1 & e^{\frac{j2\pi 3}{5}} & e^{\frac{j2\pi 2 \cdot 3}{5}} & e^{\frac{j2\pi 3 \cdot 3}{5}} & e^{\frac{j2\pi 4 \cdot 3}{5}} \\ 1 & e^{\frac{j2\pi 4}{5}} & e^{\frac{j2\pi 2 \cdot 4}{5}} & e^{\frac{j2\pi 3 \cdot 4}{5}} & e^{\frac{j2\pi 4 \cdot 4}{5}} \end{pmatrix}.$$

Spectral Analysis of Graph Data

- ▶ GSO eigendecomposition: $\mathbf{S} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}$



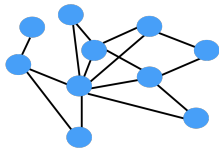
Eigendecomposition
of GSO matrix

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- ▶ $\mathbf{\Lambda} = \text{diag}([\lambda_1, \dots, \lambda_N])$: graph frequencies
- ▶ $\mathbf{V} = (\mathbf{v}_1, \dots, \mathbf{v}_N)$: graph modes (analogous to complex exponentials)
- ▶ **Graph Fourier Transform (GFT)**: $\hat{\mathbf{x}} = \mathbf{V}^{-1}\mathbf{x}$
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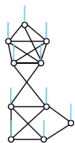
- ▶ **Q**: If we treat the eigenvectors $\mathbf{v}$ as graph modes, how to measure their variability?

- Work with (any) Laplacian matrix: $\mathbf{S} = \mathbf{L} \Rightarrow$ Symmetric (Positive Semidefinite)
 - $\Rightarrow$ Always: $\mathbf{L} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^H$
 - $\Rightarrow$ Real positive eigenvalues $\Rightarrow 0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_N$

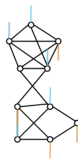
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 - $\Rightarrow$ with Laplacian quadratic form $\mathbf{x}^\top \mathbf{L} \mathbf{x}$ (Lec.2: connected nodes share similar values)
 - $\Rightarrow$ **idea:** Consider each eigenvector as graph signal $\mathbf{x} = \mathbf{v}_i$

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- ▶ **Eigenvector variability:** $\mathbf{v}_i^\top \mathbf{L} \mathbf{v}_i = \lambda_i \quad 0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_N \quad \mathbf{L} \mathbf{v}_i = \lambda_i \mathbf{v}_i$



$$\mathbf{v}_1^\top \mathbf{L} \mathbf{v}_1 = \lambda_1 = 0$$



$$\mathbf{v}_4^\top \mathbf{L} \mathbf{v}_4 = \lambda_4$$

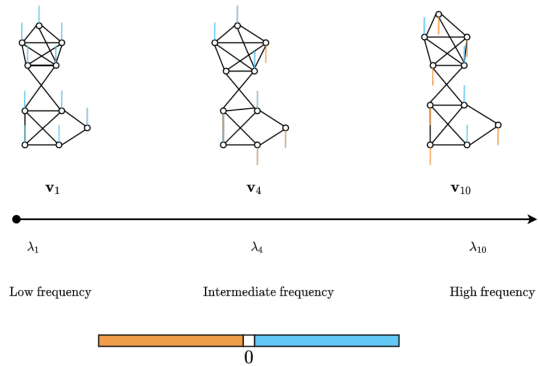
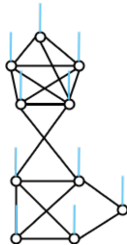


$$\mathbf{v}_{10}^\top \mathbf{L} \mathbf{v}_{10} = \lambda_{10}$$

Spectral Analysis of Graph Data

Undirected graphs

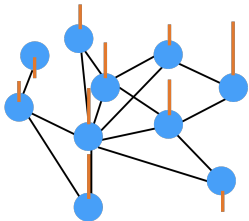
- ▶ Least variable eigenvector: $\mathbf{v}_1 = \mathbf{1}_N / \sqrt{N}$ **constant**
- ▶ Zero variability: $\mathbf{v}_1^T \mathbf{L} \mathbf{v}_1 = \mathbf{1}^T \mathbf{L} \mathbf{1} = 0$
- ▶ Eigenvalue $\lambda_1 = 0$ is the zero graph frequency



Spectral Analysis of Graph Data

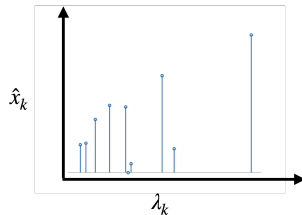
GFT for undirected graphs

- Graph Fourier Transform: $\hat{\mathbf{x}} = \mathbf{V}^H \mathbf{x}$

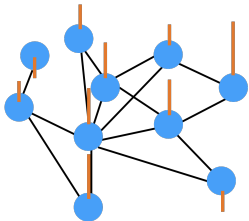


$$\hat{\mathbf{x}} = \mathbf{V}^H \mathbf{x}$$

Decompose signal

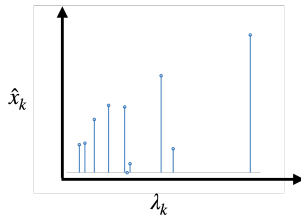


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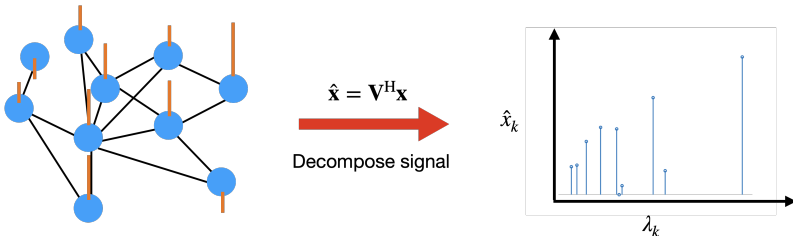
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Decompose signal



- How much each graph mode (eigenvector) contributes to the signal!

- ▶ Graph Fourier Transform: $\hat{\mathbf{x}} = \mathbf{V}^H \mathbf{x}$

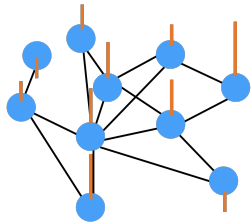


- ▶ How much each graph mode (eigenvector) contributes to the signal!
- ▶ We can now represent the data in the **graph spectral domain**
 - ⇒ Each **GFT coefficient** $\hat{x}_k$ is plotted as a bar over eigenvalue λ_k
 - ⇒ Tells how much the **variability of eigenvector** $\mathbf{v}_k$ **contributes** to the **variability of graph signal** $\mathbf{x}$

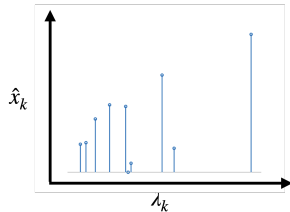
Spectral Analysis of Graph Data

Inverse GFT for undirected graphs

- Inverse GFT: $\mathbf{x} = \mathbf{V}\hat{\mathbf{x}}$



$\mathbf{x} = \mathbf{V}\hat{\mathbf{x}}$
Reconstruct signal

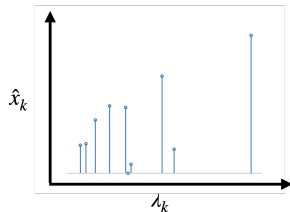
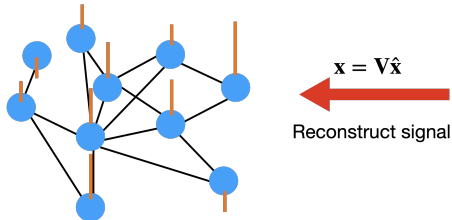


- $\mathbf{x} = \sum_{k=1}^N \hat{x}_k \mathbf{v}_k$
- Reconstruct the signal from spectral representation

Spectral Analysis of Graph Data

Inverse GFT for undirected graphs

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- $\mathbf{x} = \sum_{k=1}^N \hat{x}_k \mathbf{v}_k$
- Reconstruct the signal from spectral representation
- Q1: What type of signal has a low frequency content?
- Q2: What type of signal has a high frequency content?

Frequency Response of Graph Filters

Frequency Analysis of Graph Filters

- ▶ We can use the GFT to analyze graph filters in the frequency domain

$$\mathbf{z} = \mathbf{H}(\mathbf{S})\mathbf{x} = \sum_{k=0}^K h_k \mathbf{S}^k \mathbf{x}$$

- ▶ Substitute the GSO eigendecomposition $\mathbf{S} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}$

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- GFT of the input $\hat{\mathbf{x}} = \mathbf{V}^{-1}\mathbf{x}$

$$\mathbf{z} = \sum_{k=0}^K h_k \mathbf{V} \Lambda^k \mathbf{V}^{-1} \mathbf{x} = \sum_{k=0}^K h_k \mathbf{V} \Lambda^k \hat{\mathbf{x}}$$

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- Multiply both sides by $\mathbf{V}^{-1}$

$$\mathbf{V}^{-1} \mathbf{z} = \sum_{k=0}^K h_k \mathbf{V}^{-1} \mathbf{V} \mathbf{\Lambda}^k \hat{\mathbf{x}}$$

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$$\mathbf{z} = \sum_{k=0}^K h_k \mathbf{V} \Lambda^k \mathbf{V}^{-1} \mathbf{x} = \sum_{k=0}^K h_k \mathbf{V} \Lambda^k \hat{\mathbf{x}}$$

- Multiply both sides by $\mathbf{V}^{-1}$

$$\mathbf{V}^{-1} \mathbf{z} = \sum_{k=0}^K h_k \mathbf{V}^{-1} \mathbf{V} \Lambda^k \hat{\mathbf{x}}$$

- We have: GFT of the output $\hat{\mathbf{z}} = \mathbf{V}^{-1} \mathbf{z}$ and $\mathbf{V}^{-1} \mathbf{V} = \mathbf{I}$

$$\hat{\mathbf{z}} = \sum_{k=0}^K h_k \Lambda^k \hat{\mathbf{x}}$$

Filter Frequency Response

- Transformed the filter **input-output** relation in the spectral domain

$$\mathbf{z} = \mathbf{H}(\mathbf{S})\mathbf{x} = \sum_{k=0}^K h_k \mathbf{S}^k \mathbf{x} \quad \rightarrow \quad \hat{\mathbf{z}} = \mathbf{H}(\mathbf{\Lambda})\hat{\mathbf{x}} = \sum_{k=0}^K h_k \mathbf{\Lambda}^k \hat{\mathbf{x}}$$

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- Filter **frequency response** or **transfer function**

$$\mathbf{H}(\mathbf{\Lambda}) = \sum_{k=0}^K h_k \mathbf{\Lambda}^k = \text{diag}\left(\sum_{k=0}^K h_k \lambda_1^k, \dots, \sum_{k=0}^K h_k \lambda_N^k\right)$$

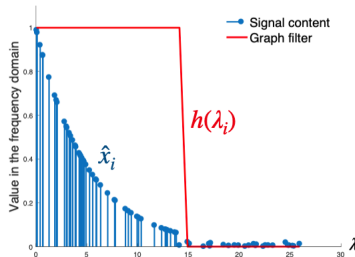
$\Rightarrow$ Diagonal matrix - polynomial in $\mathbf{\Lambda}$

Filter Frequency Response

- For a **specific graph frequency** λ_i , the filter frequency response is

$$\hat{z}_i = h(\lambda_i)\hat{x}_i = \left(\sum_{k=0}^K h_k \lambda_i^k \right) \hat{x}_i$$

Point-wise multiplication

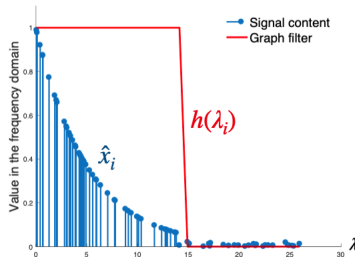


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Point-wise multiplication



- **Learning** the filter parameters h_k implies
 - ⇒ **Learning** the frequency response
 - ⇒ **Learning** how to alter the spectral content of the graph signal (node features)

Convolution Theorem for Graphs

- Convolution principle:
 - ⇒ Shift and sum operation in the nominal domain (time/space/vertex)
 - ⇒ Point-wise multiplication in the spectral domain (Fourier)

Convolution Theorem for Graphs

- Convolution principle:

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- The filter respects the convolution theorem

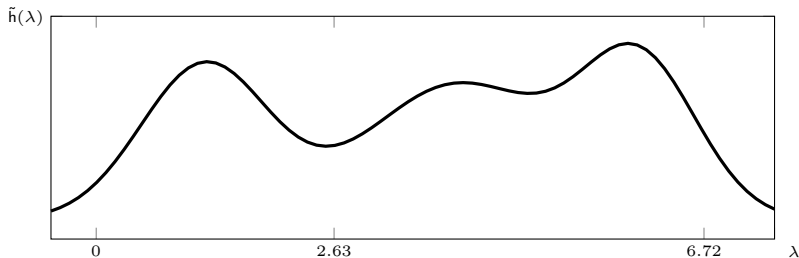
$$\mathbf{H}(\mathbf{S})\mathbf{x} = \sum_{k=0}^K h_k \mathbf{S}^k \mathbf{x} \quad \rightarrow \quad \mathbf{H}(\mathbf{\Lambda})\hat{\mathbf{x}} = \sum_{k=0}^K h_k \mathbf{\Lambda}^k \hat{\mathbf{x}}$$

- ⇒ Shift-and-sum in the vertex domain

- ⇒ Pointwise multiplication in the spectral domain

Effect of Running the Same Filter on Different Graphs

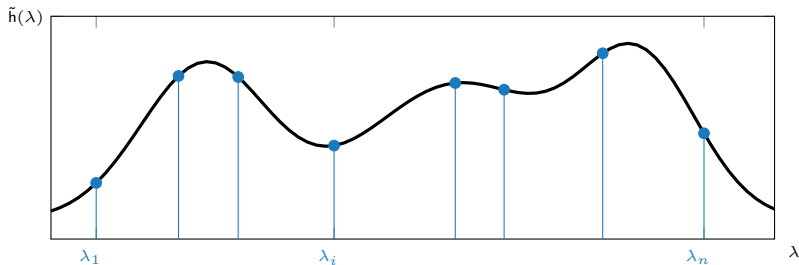
- Given coefficients $\mathbf{h} = \{h_k\}$ the graph filter frequency response is determined $H(\lambda) = \sum_{k=0}^K h_k \lambda^k$
- ⇒ Universal for any graph



Slide credit: F. Gama

Effect of Running the Same Filter on Different Graphs

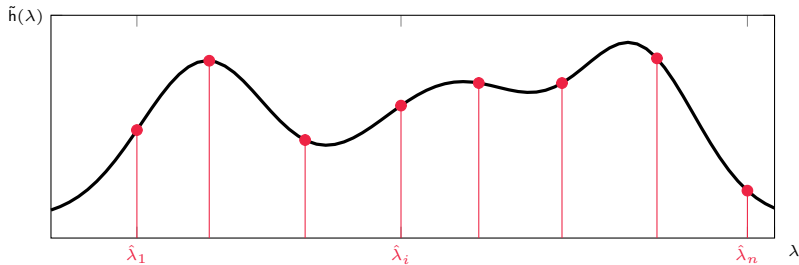
- ▶ Given coefficients $\mathbf{h} = \{h_k\}$ the graph filter frequency response is determined $H(\lambda) = \sum_{k=0}^K h_k \lambda^k$
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- ▶ For a **specific graph**, the response is instantiated on its **specific eigenvalues** λ_i



Slide credit: F. Gama

Effect of Running the Same Filter on Different Graphs

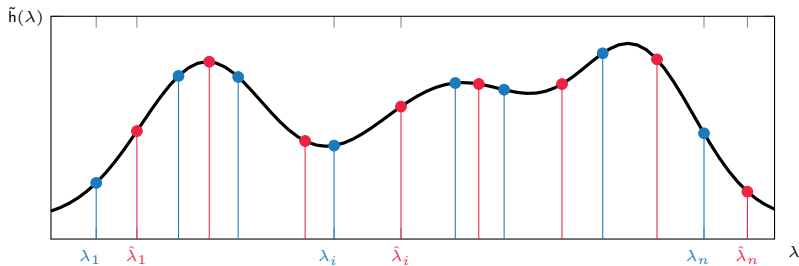
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- ▶ For a **different graph**, the response is instantiated **different eigenvalues** λ_i



Slide credit: F. Gama

Effect of Running the Same Filter on Different Graphs

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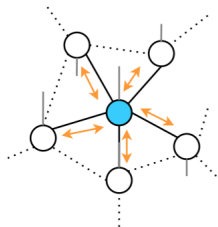
Slide credit: F. Gama

Filter Spectral Locality

- ▶ Filter frequency response is polynomial $H(\lambda) = \sum_{k=0}^K h_k \lambda^k$
- ▶ Polynomial order defines the filter locality

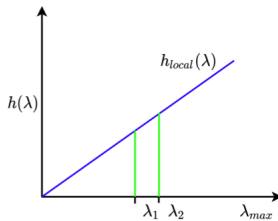
Filter Spectral Locality

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$$H(S) = h_0 + h_1 S$$

Order-1 filter
Local in the neighborhood

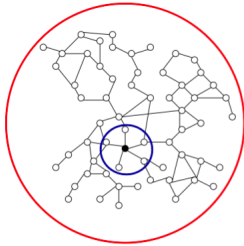


$$H(\lambda) = h_0 + h_1 \lambda$$

Order-1 filter
Non-local in the spectrum

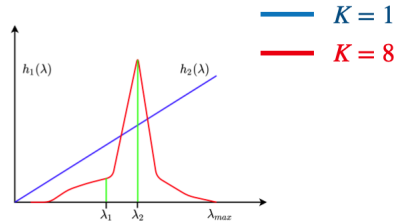
Filter Spectral Locality

cont.



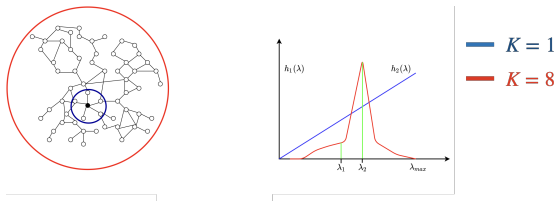
Order-8 filter
Non-local in the neighborhood

$$H(S) = \sum_{k=0}^8 h_k S^k$$



$$H(\lambda) = \sum_{k=0}^8 h_k \lambda^k$$

Order-8
Local in the spectrum ..



- ▶ **Graph convolutional filters** can be local either in the vertex or in the spectral domain
- ▶ **Trade-off:**
 - ⇒ **Low-order:** less interactions with neighbors (distributed !); limited response
 - ⇒ **High-order:** more interactions with neighbors (distributed !); stronger response

Learning Activity: Interpreting Graph Frequencies

The eigenvectors of the Laplacian act as graph Fourier modes. Their associated eigenvalues quantify how much the mode varies across edges. This gives meaning to the concepts of low-frequency and high-frequency graph signals.

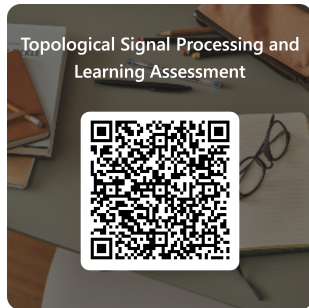
- ▶ **Q10.1:** Consider an undirected graph with Laplacian $\mathbf{L} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^\top$, where $0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_N$. Which statement is correct?

- ▶ **Instructions**

- ⇒ Scan the QR code to open the activity
- ⇒ Work individually first
- ⇒ Then discuss your answer in pairs or groups of 3
- ⇒ Submit your final answer in Microsoft Forms

- ▶ **Logistics**

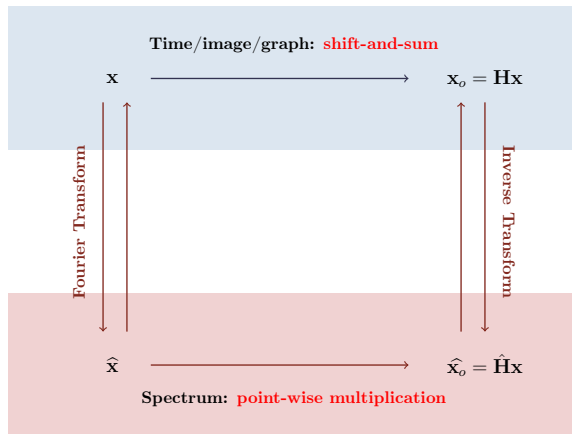
- ⇒ **Time:** 8 min
- ⇒ **Goal:** Build intuition on graph frequencies and smoothness
- ⇒ **Product:** One submitted response per student + short justification to peers



<https://forms.office.com/>

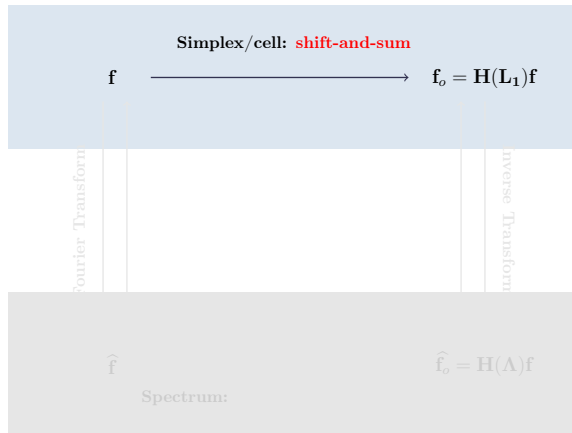
Spectral Analysis beyond Graphs

Convolution Theorem



- Convolution theorem as a principled way to generalize concepts to new domains

Topological Shift-and-Sum



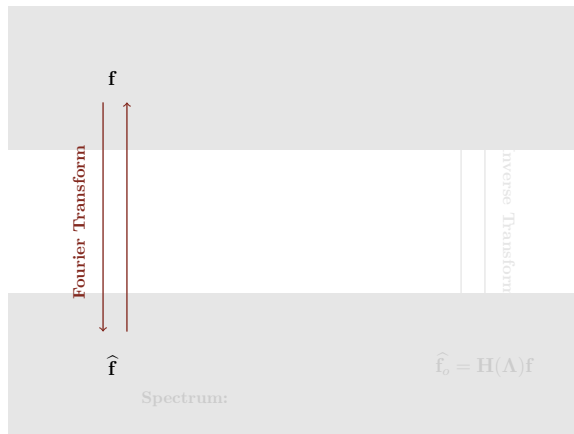
- We start with the **shift-and-sum** principle

Definition. Simplicial convolutional filters

$$\mathbf{H}(\mathbf{L}_1) = h\mathbf{I} + \sum_{l_1=1}^{L_1} h_{l,l_1} (\mathbf{L}_{1,\ell})^{l_1} + \sum_{l_2=1}^{L_2} h_{u,l_2} (\mathbf{L}_{1,u})^{l_2}$$

- ▶ Filter output: $\mathbf{f}_o = \mathbf{H}(\mathbf{L}_1)\mathbf{f}$
- ▶ Signals are shifted over lower and upper neighbors

Topological Shift-and-Sum



- Topological **Fourier** transform ensures spectral **processing** of topological **signals**

Hodge Decomposition

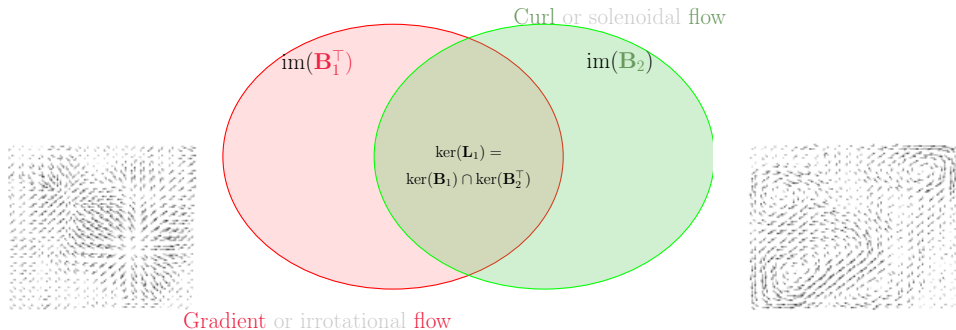
- ▶ Hodge Laplacian structure $\mathbf{L}_1 = \mathbf{B}_1^\top \mathbf{B}_1 + \mathbf{B}_2 \mathbf{B}_2^\top$
 - ⇒ The space of edge signals can be decomposed into three subspaces

$$\mathbb{R}^{N_1} \equiv \underbrace{\text{im}(\mathbf{B}_1^\top)}_{\text{Gradient}} \oplus \underbrace{\ker(\mathbf{L}_1)}_{\text{Harmonic}} \oplus \underbrace{\text{im}(\mathbf{B}_2)}_{\text{curl}}$$

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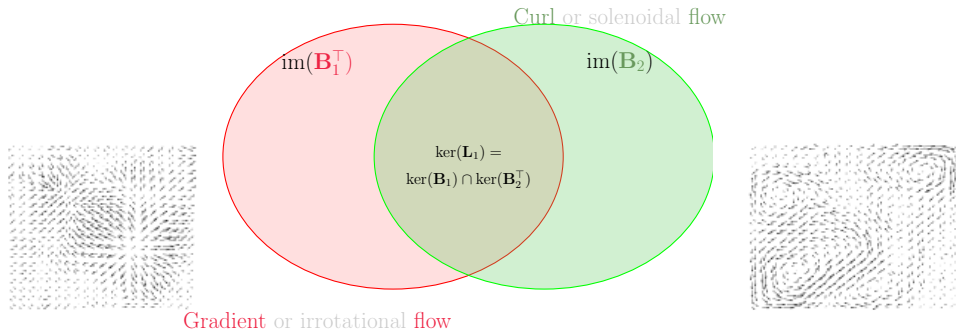
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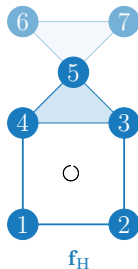
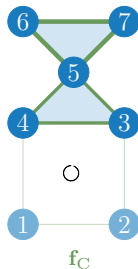
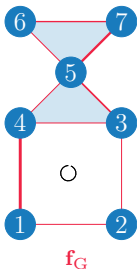
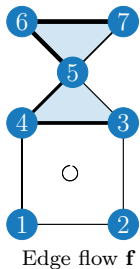
- ▶ The dimensions of $\ker(\mathbf{L}_1)$ are **topological invariants** for k -simplicial complex no. of holes, cavities

Hodge Decomposition

$$\mathbb{R}^{N_1} \equiv \underbrace{\text{im}(\mathbf{B}_1^T)}_{\text{Gradient}} \oplus \underbrace{\ker(\mathbf{L}_1)}_{\text{Harmonic}} \oplus \underbrace{\text{im}(\mathbf{B}_2)}_{\text{curl}}$$

- An edge flow signal can be decomposed as

$$\mathbf{f} = \mathbf{f}_G + \mathbf{f}_C + \mathbf{f}_H$$

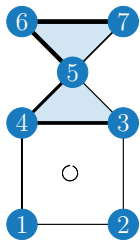


Hodge Decomposition

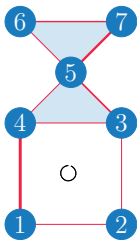
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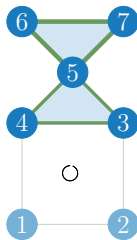
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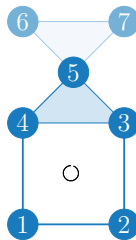
Edge flow $\mathbf{f}$



$\mathbf{f}_G$



$\mathbf{f}_C$



$\mathbf{f}_H$

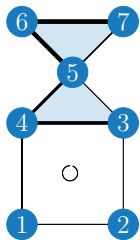
⇒ Gradient flow $\mathbf{f}_G := \mathbf{B}_1^T \mathbf{v}$ difference of node signals, zero flow-sum along triangles

Hodge Decomposition

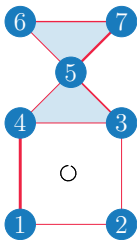
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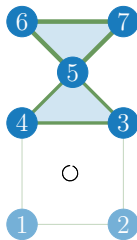
$$\mathbf{f} = \mathbf{f}_G + \mathbf{f}_C + \mathbf{f}_H$$



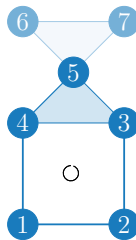
Edge flow $\mathbf{f}$



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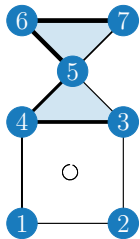
⇒ Curl flow $\mathbf{f}_C := \mathbf{B}_2 \mathbf{t}$ local flows circulating around triangles, in-flow on each node equals out-flow

Hodge Decomposition

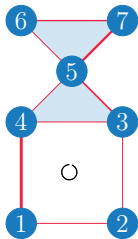
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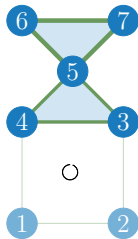
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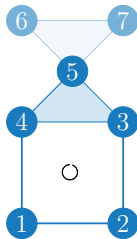
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⇒ Harmonic flow $\mathbf{f}_H$: global circulating flow

Spectral Approach to Hodge Decomposition

- ▶ The Hodge decomposition is a consequence of the Hodge Laplacian structure $\Rightarrow \mathbf{L}_1 = \mathbf{B}_1^\top \mathbf{B}_1 + \mathbf{B}_2 \mathbf{B}_2^\top$

$$\mathbb{R}^{N_1} \equiv \underset{\text{Gradient}}{\text{im}(\mathbf{B}_1^\top)} \oplus \underset{\text{Harmonic}}{\text{ker}(\mathbf{L}_1)} \oplus \underset{\text{curl}}{\text{im}(\mathbf{B}_2)}$$

- ▶ Take the eigendecomposition $\mathbf{L}_1 = \mathbf{U} \mathbf{\Lambda} \mathbf{U}^\top$

$\Rightarrow$ The eigenvectors $\mathbf{U}$ can be rearranged into $\mathbf{U} = [\mathbf{U}_G \ \mathbf{U}_C \ \mathbf{U}_H]$

$\Rightarrow$ The eigenvalues $\mathbf{\Lambda}$ can be collected as $\mathbf{\Lambda} = \text{blkdiag}(\mathbf{\Lambda}_G, \mathbf{\Lambda}_C, \mathbf{\Lambda}_H = \mathbf{0})$

- ▶ **Gradient** eigenvectors $\mathbf{U}_G$: the eigenvectors of $\mathbf{L}_{1,\ell}$ corresponding to the eigenvalues $\lambda_{G,i} > 0$.
- ▶ **Curl** eigenvectors $\mathbf{U}_C$: the eigenvectors of $\mathbf{L}_{1,u}$ corresponding to the eigenvalues $\lambda_{C,i} > 0$.
- ▶ **Harmonic** eigenvectors $\mathbf{U}_H$: the eigenvectors of $\mathbf{L}_1$ corresponding to the zero eigenvalues $\lambda_{H,i} = 0$.

Topological Spectral Dictionaries

- ▶ Edge flow signal can be decomposed as

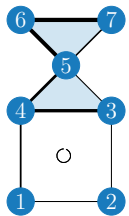
$$\mathbf{f} = \mathbf{f}_G + \mathbf{f}_C + \mathbf{f}_H$$

- ▶ Hodge eigenvectors:

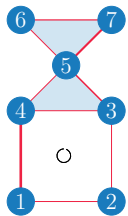
⇒ $\mathbf{U}_G$ spans the **gradient** subspace, $\mathbf{f}_G = \mathbf{U}_G \hat{\mathbf{f}}_G$

⇒ $\mathbf{U}_C$ spans the **curl** subspace, $\mathbf{f}_C = \mathbf{U}_C \hat{\mathbf{f}}_C$

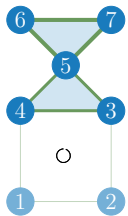
⇒ $\mathbf{U}_H$ spans the **harmonic** subspace, $\mathbf{f}_H = \mathbf{U}_H \hat{\mathbf{f}}_H$



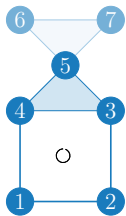
Edge flow $\mathbf{f}$



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$\mathbf{f}_C$



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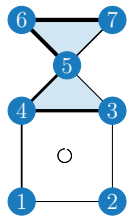
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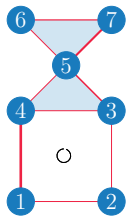
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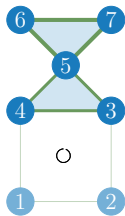
⇒ $\mathbf{U}_H$ spans the **harmonic** subspace, $\mathbf{f}_H = \mathbf{U}_H \hat{\mathbf{f}}_H$



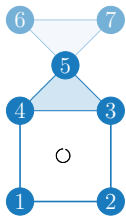
Edge flow $\mathbf{f}$



$\mathbf{f}_G$



$\mathbf{f}_C$



$\mathbf{f}_H$

- ▶ Q: Do these **eigenvectors** carry a notion of topological **variability**?

Eigenvector Variability

- Consider the quadratic form

$$g(\mathbf{f}) = \mathbf{f}^\top \mathbf{L}_{1,u} \mathbf{f} = \mathbf{f}^\top \mathbf{B}_2 \mathbf{B}_2^\top \mathbf{f} = \|\mathbf{B}_2^\top \hat{\mathbf{f}}\|_2^2$$

- Measures "smoothness: w.r.t. the upper Laplacian $\mathbf{L}_{1,u} = \mathbf{B}_2 \mathbf{B}_2^\top$
 $\Rightarrow$ flows with minimum quadratic form minimize the total curl

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- Measures "smoothness: w.r.t. the upper Laplacian $\mathbf{L}_{1,u} = \mathbf{B}_2 \mathbf{B}_2^\top$
 $\Rightarrow$ flows with minimum quadratic form minimize the total curl
- The eigenvectors of the upper Laplacian $\mathbf{L}_{1,u}$
 $\Rightarrow$ curl eigenvectors $\mathbf{U}_C = [\mathbf{u}_{C,1}, \dots, \mathbf{u}_{C,N_C}]$ corresponding to the eigenvalues $\lambda_{C,i} > 0$

Eigenvector Variability

- Consider the quadratic form

$$g(\mathbf{f}) = \mathbf{f}^\top \mathbf{L}_{1,u} \mathbf{f} = \mathbf{f}^\top \mathbf{B}_2 \mathbf{B}_2^\top \mathbf{f} = \|\mathbf{B}_2^\top \hat{\mathbf{f}}\|_2^2$$

- Measures "smoothness": w.r.t. the upper Laplacian $\mathbf{L}_{1,u} = \mathbf{B}_2 \mathbf{B}_2^\top$
 $\Rightarrow$ flows with minimum quadratic form minimize the total curl
- The eigenvectors of the upper Laplacian $\mathbf{L}_{1,u}$
 $\Rightarrow$ curl eigenvectors $\mathbf{U}_C = [\mathbf{u}_{C,1}, \dots, \mathbf{u}_{C,N_C}]$ corresponding to the eigenvalues $\lambda_{C,i} > 0$
- Thus, using $g(\cdot)$ we can sort the curl eigenvectors based on their total curl

$$g(\mathbf{u}_{C,i}) = \lambda_{C,i} \leq g(\mathbf{u}_{C,j}) = \lambda_{C,j}$$

Eigenvector Variability

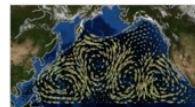
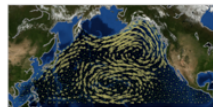
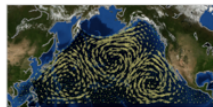
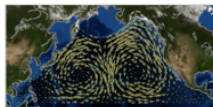
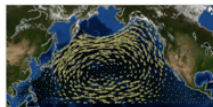
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$$g(\mathbf{u}_{C,i}) = \lambda_{C,i} \leq g(\mathbf{u}_{C,j}) = \lambda_{C,j}$$

- ▶ Eigenvalues $\lambda_{C,i}$ are the $\mathbf{L}_{1,u}$ curl frequencies



λ_C , more rotational

Eigenvector Variability

► Hodge Laplacian $\mathbf{L}_1 = \mathbf{B}_1^\top \mathbf{B}_1 + \mathbf{B}_2 \mathbf{B}_2^\top$

⇒ measure also the variability of the lower Laplacian eigenvectors $\mathbf{L}_{1,\ell} = \mathbf{B}_1^\top \mathbf{B}_1$

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⇒ solution measure the total divergence

Eigenvector Variability

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- ▶ The eigenvectors of the lower Laplacian $\mathbf{L}_{1,\ell}$

⇒ curl eigenvectors $\mathbf{U}_G = [\mathbf{u}_{G,1}, \dots, \mathbf{u}_{G,N_G}]$ corresponding to the eigenvalues $\lambda_{G,i} > 0$

Eigenvector Variability

- ▶ The eigenvectors of the lower Laplacian $\mathbf{L}_{1,\ell}$ can be grouped into
 - $\Rightarrow$ eigenvectors corresponding to eigenvalues $\lambda_i = 0$
 - $\Rightarrow$ curl eigenvectors $\mathbf{U}_G = [\mathbf{u}_{G,1}, \dots, \mathbf{u}_{G,N_G}]$ corresponding to the eigenvalues $\lambda_{G,i} > 0$
- ▶ Thus, using $g(\cdot)$ we can sort the gradient eigenvectors based on their total divergence

$$g(\mathbf{u}_{G,i}) = \lambda_{G,i} \leq g(\mathbf{u}_{G,j}) = \lambda_{G,j}$$

- ▶ Eigenvalues $\lambda_{G,i}$ are the **gradient** frequencies



Topological Fourier Transform

- We can build unitary spectral dictionaries for signals of any order using the eigenvectors

$$\mathbf{U} = [\mathbf{U}_G \mathbf{U}_C \mathbf{U}_H]$$

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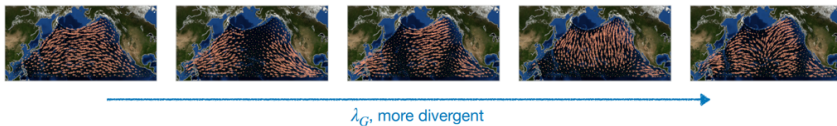
- ▶ The **TFT** measures how much the **variability** gradient/curl of each respective **eigenvector** contributes towards the *variability* gradient/curl of the flow
- ▶ Inverse topological Fourier transform $\mathbf{f} = \mathbf{U} \hat{\mathbf{f}}$
- ▶ The GFT and DFT are special cases of the topological Fourier transform

Topological Fourier Transform

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Topological Fourier Transform

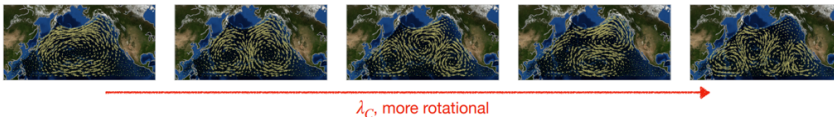
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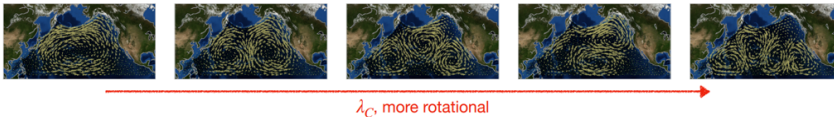
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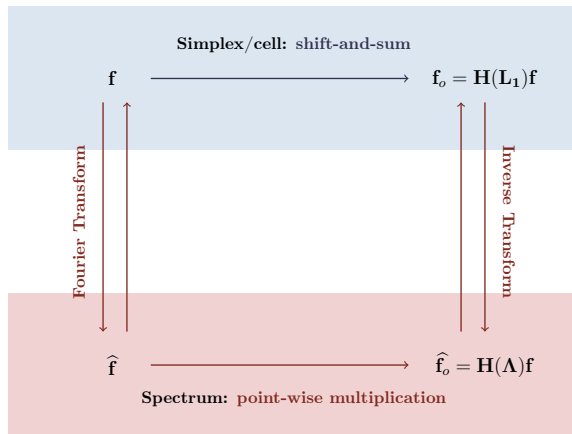
- Eigenvalues $\lambda_{G,i}$ are the **gradient** frequencies



- Eigenvalues $\lambda_{C,i}$ are the **curl** frequencies



- Eigenvalues $\lambda_{H,i=0}$ are the **harmonic** frequencies



- We can now do spectral analysis and filtering of topological signals

Frequency Response of Simplicial Convolutional Filter

$$\mathbf{H}(\mathbf{L}_1) = \mathbf{h}\mathbf{I} + \sum_{l_1=1}^{L_1} h_{l,l_1} (\mathbf{L}_{1,\ell})^{l_1} + \sum_{l_2=1}^{L_2} h_{u,l_2} (\mathbf{L}_{1,u})^{l_2}$$

- Q: How does the filter behave in the Hodge spectral domain?

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- **Q:** How does the **filter** behave in the Hodge **spectral** domain?
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$$\mathbf{L}_1 = \mathbf{U} \text{blkdiag}(\mathbf{\Lambda}_H = \mathbf{0}, \mathbf{\Lambda}_G, \mathbf{\Lambda}_C) \mathbf{U}^\top$$

$$\Rightarrow \mathbf{L}_{1,\ell} = \mathbf{U} \text{blkdiag}(\mathbf{0}, \mathbf{\Lambda}_G, \mathbf{0}) \mathbf{U}^\top \quad \text{and} \quad \mathbf{L}_{1,u} = \mathbf{U} \text{blkdiag}(\mathbf{0}, \mathbf{0}, \mathbf{\Lambda}_C) \mathbf{U}^\top$$

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- Independent **pointwise multiplication** control of the gradient and curl embeddings

$$\mathbf{f}_o = \mathbf{H}(\mathbf{L}_1)\mathbf{f} \iff \hat{\mathbf{f}}_o = \mathbf{H}(\mathbf{\Lambda})\hat{\mathbf{f}}$$

Convolution in the Spectrum

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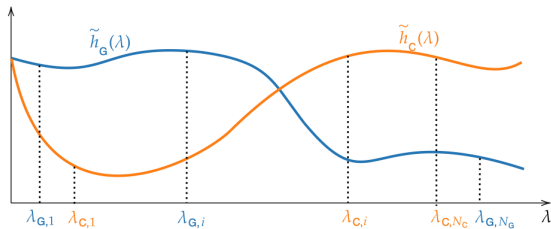
$$\begin{cases} \hat{\mathbf{f}}_G = \mathbf{U}_G^\top \mathbf{f}, & \text{gradient embedding} \\ \hat{\mathbf{f}}_C = \mathbf{U}_C^\top \mathbf{f}, & \text{curl embedding} \\ \hat{\mathbf{f}}_H = \mathbf{U}_H^\top \mathbf{f}, & \text{harmonic embedding} \end{cases} \implies \begin{cases} \hat{\mathbf{f}}_{G,o} = \left(h\mathbf{I} + \sum_{l_1=1}^{L_1} h_{l,l_1} \mathbf{\Lambda}_G^{l_1} \right) \hat{\mathbf{f}}_G \\ \hat{\mathbf{f}}_{C,o} = \left(h\mathbf{I} + \sum_{l_2=1}^{L_2} h_{u,l_2} \mathbf{\Lambda}_C^{l_2} \right) \hat{\mathbf{f}}_C \\ \hat{\mathbf{f}}_{H,o} = h\hat{\mathbf{f}}_H \end{cases}$$

Convolution in the Spectrum

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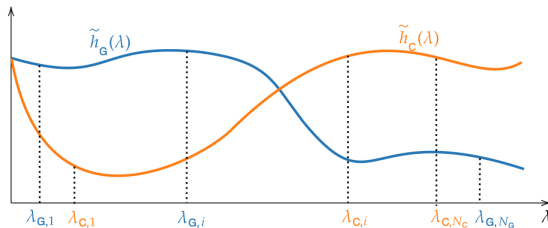
- The **spectral responses** act **independently** on the respective embeddings
- Coupled at the harmonic component

- ▶ Consider an SCNN trained on simplex $\mathbf{L}_1 = \Phi(\mathbf{F}, \mathbf{L}_1, \mathcal{H})$
- ▶ And deployed on the perturbed simplex $\tilde{\mathbf{L}}_1 = \Phi(\mathbf{F}, \tilde{\mathbf{L}}_1, \mathcal{H})$

Theorem (Stability). For

1. small relative perturbations $\|\mathbf{L}_1 - \tilde{\mathbf{L}}_1\| \leq \epsilon$
 2. bounded variability of both gradient and curl frequency responses $|h(\lambda_i) - h(\lambda_j)| \leq H|\lambda_i - \lambda_j|$
- the SCNN output can be bounded as

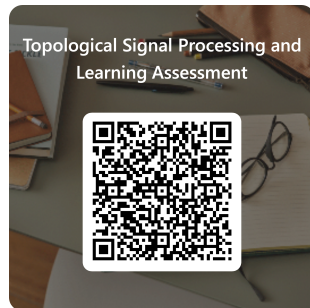
$$\|\Phi(\mathbf{F}, \mathbf{L}_1, \mathcal{H}) - \Phi(\mathbf{F}, \tilde{\mathbf{L}}_1, \mathcal{H})\| \leq O(LF^L H \|\mathbf{F}\| \epsilon).$$



Learning Activity: Frequency Response of Simplicial Filters

Simplicial convolutional filters generalize graph filters by acting separately on the **gradient** and **curl** subspaces. In the Hodge spectral domain, the filter response becomes diagonal and acts independently on the gradient and curl embeddings.

- ▶ **Q10.2:** A simplicial convolutional filter is designed so that it strongly attenuates high curl frequencies but leaves gradient frequencies mostly unchanged. What is the most direct effect on an edge flow signal?
- ▶ **Instructions**
 - ⇒ Scan the QR code to open the activity
 - ⇒ Work individually first
 - ⇒ Then discuss your answer in pairs or groups of 3
 - ⇒ Submit your final answer in Microsoft Forms
- ▶ **Logistics**
 - ⇒ **Time:** 10 min
 - ⇒ **Goal:** Link spectral filter design to manipulation of flow components
 - ⇒ **Product:** One submitted response per student + reasoning shared in group



<https://forms.office.com/>

Further Reading

- ▶ (Isufi et al) Graph filters for processing and learning from network data
⇒ Ch. 4
- ▶ (Isufi et al) Graph filters for signal processing and machine learning on graphs
⇒ Sec. IV
- ▶ (Shuman et al) The emerging field of signal processing on graphs: Extending high-dimensional data analysis to networks and other irregular domains
⇒ Until convolution (excluding)
- ▶ (Ortega et al) Graph Signal Processing: Overview, Challenges, and Applications
⇒ Sec. I and II
- ▶ (Yang et al) "Simplicial Convolutional Filters", IEEE TSP 2020
⇒ extends spectral analysis to the topological domain
- ▶ Barbarossa, S., and Sardellitti, S. (2020). Topological signal processing over simplicial complexes. IEEE Transactions on Signal Processing, 68, 2992-3007.
- ▶ Schaub, Michael T., Yu Zhu, Jean-Baptiste Seby, T. Mitchell Roddenberry, and Santiago Segarra. "Signal processing on higher-order networks: Livin'on the edge... and beyond." Signal Processing 187 (2021): 108149.

Video Material

- ▶ *Task-driven Topology Inference for Signal processing and Learning over Topological Domains*
⇒ <https://www.youtube.com/watch?v=PPy8SJpmkCU>
- ▶ *Topological signal processing and learning* <https://www.youtube.com/watch?v=Ylwz6BWeIvA>